



Foundation **SERIES**

Chemistry

THERMOCHEMISTRY



THERMOCHEMISTRY



It is the branch of physical chemistry which deals with the heat change during a chemical reaction or during a phase change.



Heat change is studied mainly in terms of enthalpy change, here we are not interested in absolute enthalpy.



In thermochemistry we are just interested in depicting energy changes, but we do not take into consideration about the thermodynamic feasibility of the chemical reaction.



STANDARD CONDITION



For gas → Pressure 1 Bar $\cong$ 1 atm

Temp = 298.15 K



For solid /Liq. → Solid & Liquid in their pure form at 1 bar pressure.



All type of enthalpies are studied for 1 mole.



STANDARD STATE OF ELEMENT



Standard state of all metals is, considered to be solid except Hg(l).

Example : Mg(s), Fe(s), Na(s)



Standard state of noble gases, N₂, H₂, O₂, F₂, Cl₂ is, considered to be gaseous state.

— Br₂(l), I₂(s)

— C(s), Graphite

— S₈(s), Rhombic

— P₄(s), White

— Sn(s), White

— As(s), Grey

— H⁺(aq.) — Standard state by convention



THERMO-CHEMICAL EQUATION

State

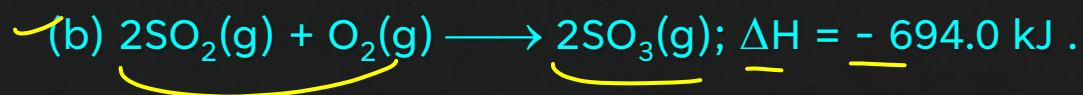
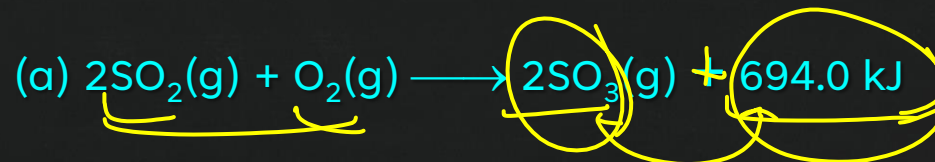
Enthalpy data



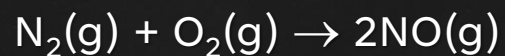
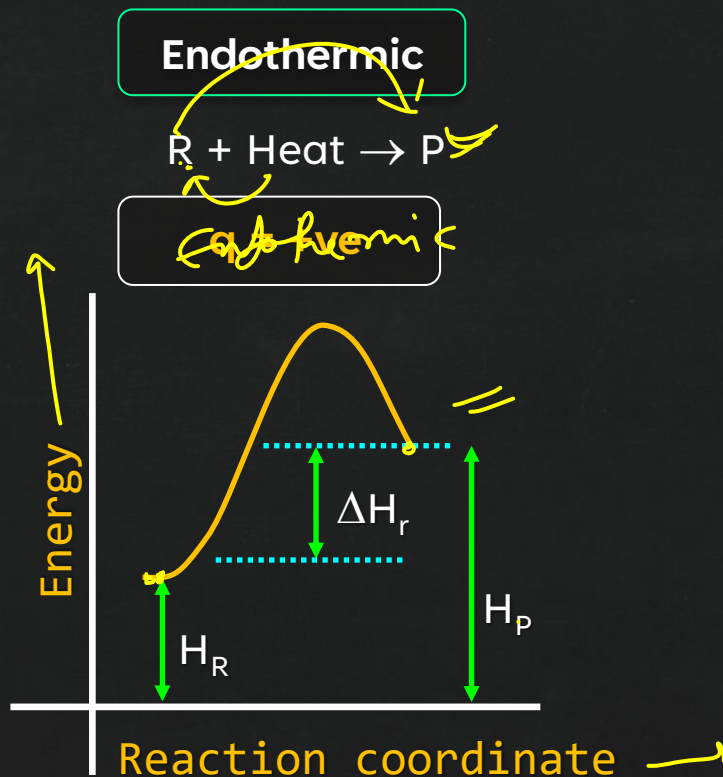
A thermochemical equation may be defined as a chemical equation which indicates the amount of heat evolved or absorbed during the chemical reaction in addition to the quantities of reactants or products.



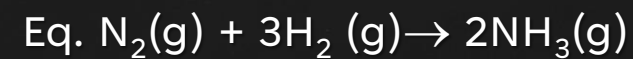
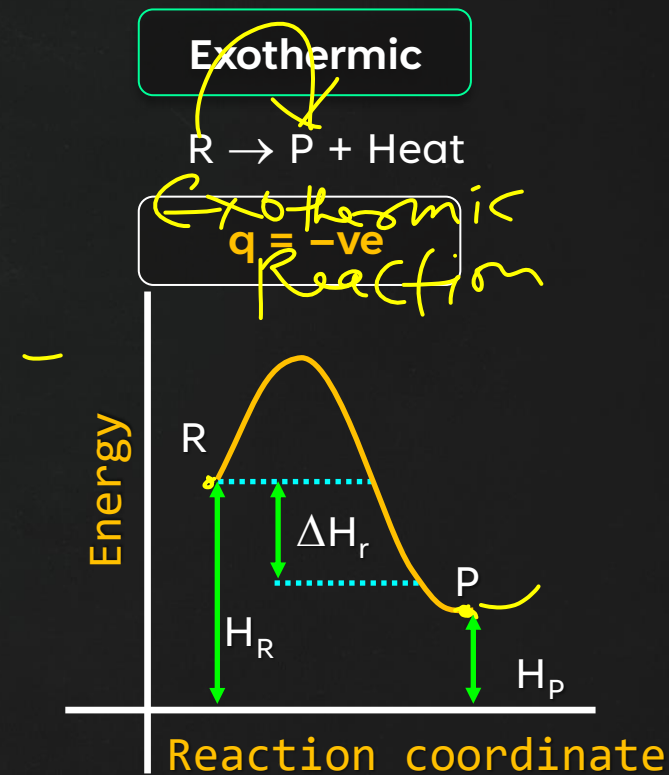
A thermochemical equation can be written in two ways :



Types of reaction (According to Heat)



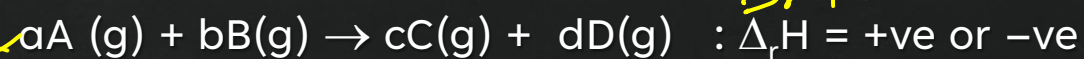
$$\Delta H_{\text{Rxn}} = 180.81 \text{ kJ}$$



$$\Delta H_{\text{Rxn}} = -92 \text{ kJ}$$



ENTHALPY CHANGE OF REACTION/HEAT OF REACTION



Enthalpy of reaction is always written as:-

$$\Delta_r H = (\sum H)_P - (\sum H)_R$$

$$\Delta_r H = (c \cdot H_C + d \cdot H_D) - (a \cdot H_A + b \cdot H_B)$$

Standard Enthalpy

$$\Delta_r H^\circ = (c \cdot H_C^\circ + d \cdot H_D^\circ) - (a \cdot H_A^\circ + b \cdot H_B^\circ)$$

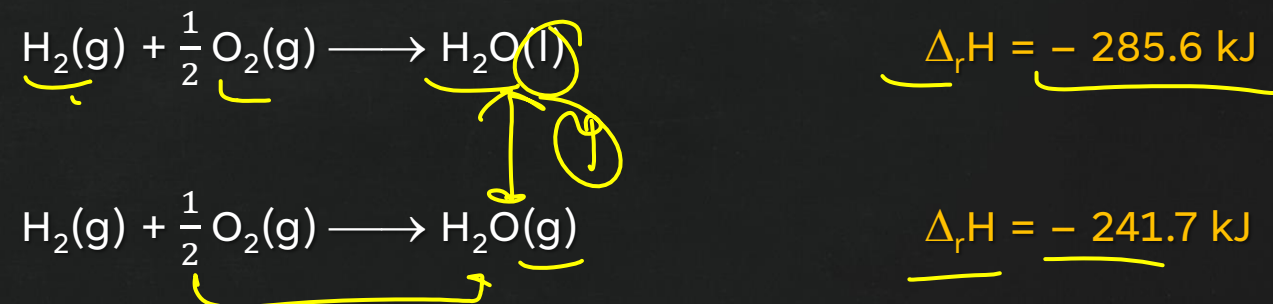
But in case of bond enthalpy/enthalpy of combustion, it is written as :-

$$\Delta_r H = (\sum H)_R - (\sum H)_P$$



FACTORS AFFECTING ENTHALPY CHANGE OF REACTION

1. Physical state of reactants and products

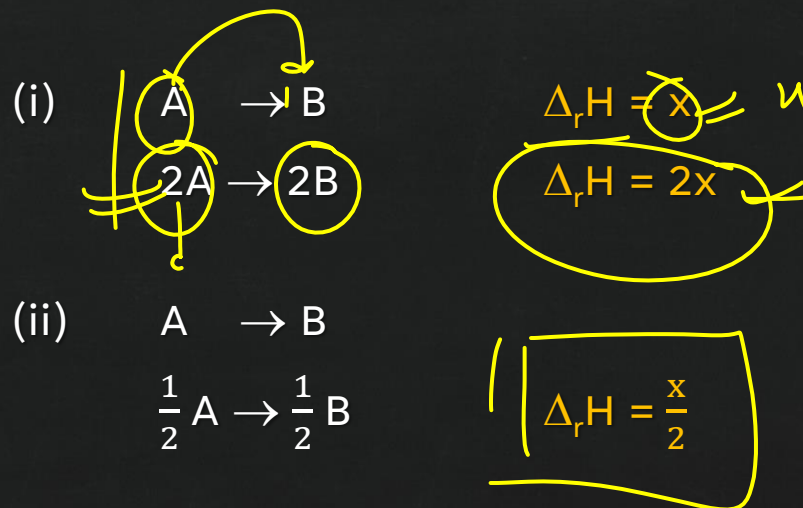


Physical state of the substance is therefore always represented in every **THERMOCHEMICAL** equation.



FACTORS AFFECTING ENTHALPY OF REACTION

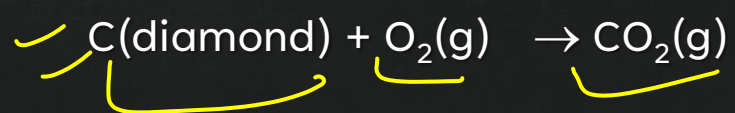
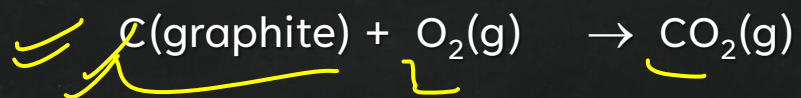
2. Stoichiometry of reaction





FACTORS AFFECTING ENTHALPY OF REACTION

3. Allotropic form



$$\Delta H_1 > \Delta H_2$$

$$\Delta H_1 = -x$$

$$\Delta H_2 = -y$$



Graphite is more thermodynamic stable than diamond.

$$\Delta H = \Delta U + \Delta n_g RT$$



FACTORS AFFECTING ENTHALPY OF REACTION

$$q_p = q_v + \Delta n_g RT$$

4. Reaction carried out at constant pressure or constant volume

$$\Delta U + \Delta n_g RT = \Delta H$$

$$\text{or, } q_v + \Delta n_g RT = q_p$$

q_v = Heat change at constant volume

q_p = Heat change at constant pressure

$$\Delta n_g = \left[\begin{array}{c} \text{Total number of moles} \\ \text{of gaseous products} \end{array} \right] - \left[\begin{array}{c} \text{Total number of moles} \\ \text{of gaseous reactants.} \end{array} \right]$$

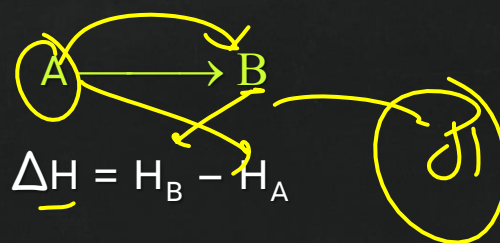


TEMPERATURE EFFECT / KIRCHHOFF'S RELATION



Effect of temp. on enthalpy of reaction can be explained by Kirchhoff's equation.

let us consider a reaction occurring at constant pressure



$$\Delta H = \int n C_p dT$$

$$\left(\frac{dH}{dT} \right)_P = C_p$$

$$\left(\frac{d(\Delta H)}{dT} \right)_P = \left(\frac{dH_B}{dT} \right)_P - \left(\frac{dH_A}{dT} \right)_P = \underline{C_{p,B}} - \underline{C_{p,A}}$$

$$\frac{d\Delta H}{dT} = \frac{C_{p,B} - C_{p,A}}{\Delta C_{p,r}} \Rightarrow \boxed{d\Delta H = \Delta C_p \cdot dT}$$

$$d(\Delta H) = \Delta C_p dT$$

Where , $\Delta C_p = \sum (C_p)_B - \sum (C_p)_A$

Similarly,

$$d(\Delta U) = \Delta C_v dT$$

Where , $\Delta C_v = \sum (C_v)_B - \sum (C_v)_A$

Case 1

If ΔC_p , ΔC_v are temperature dependent.

$C_p/C_v \rightarrow$ ① fn of T
 $\rightarrow$ ② independent of T

$$\int d(\Delta H) = \int_{T_1}^{T_2} \Delta C_p dT$$

$$\therefore \Delta H_{T_2} = \Delta H_{T_1} + \int_{T_1}^{T_2} \Delta C_p dT$$

Similarly, $\int d(\Delta U) = \int_{T_1}^{T_2} \Delta C_V dT$

$$\therefore \Delta U_{T_2} = \Delta U_{T_1} + \int_{T_1}^{T_2} \Delta C_V dT$$

Case 2

ΔH

If ΔC_p , ΔC_v are constant between two temperatures T_1 & T_2 .

$$\Delta H_{T_2} - \Delta H_{T_1} = \Delta C_p (T_2 - T_1) \quad \text{----- (i)}$$

Similarly, $\Delta U_{T_2} - \Delta U_{T_1} = \Delta C_v (T_2 - T_1) \quad \text{----- (ii)}$

Equations (i) & (ii) are known as Kirchoff's equations

Example

$$= 2.5 \left[16.5 \times 100 + \frac{10^{-2}}{2} (160000 - 90000) \right]$$

$$= 2.5 \left(1650 + \frac{10^{-2}}{2} \cdot 70000 \right)$$

$$= 2.5 (1650 + 350) = 5000 \text{ J} = \underline{5 \text{ kJ}}$$

The constant volume molar heat capacity of an ideal gas is expressed by $C_{v,m} = 16.5 + 10^{-2} T$ (All values are in SI units). If 2.5 mole of this gas at constant volume is heated from 27°C to 127°C , the internal energy increases by "x" kJ. Hence, x is

Solution

$$\Delta U = \int_{T_1}^{T_2} n C_{v,m} dT$$

$$= 2.5 \int_{T_1}^{T_2} (16.5 + 10^{-2} T) dT$$

$$= 2.5 \left[16.5 (T_2 - T_1) + \frac{10^{-2}}{2} (T_2^2 - T_1^2) \right]$$

Ans: 5



HESS'S LAW OF CONSTANT HEAT SUMMATION



Enthalpy change in, a given chemical or physical process is the same whether the process is carried out in one step or in several steps.

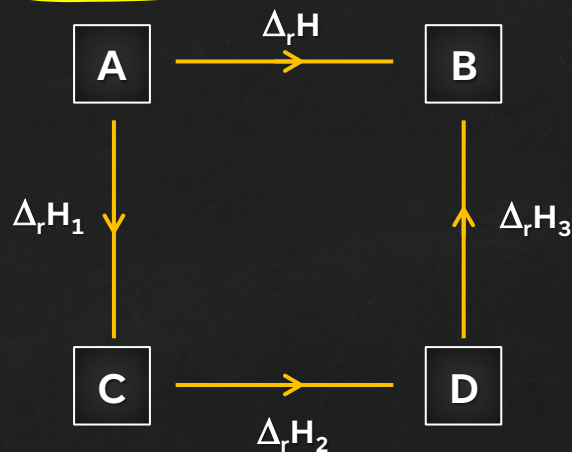


Enthalpy change in, a given chemical or physical process is independent from intermediate steps and depends on initial & final state only.



Enthalpy change is independent of the time consumed in the reaction.

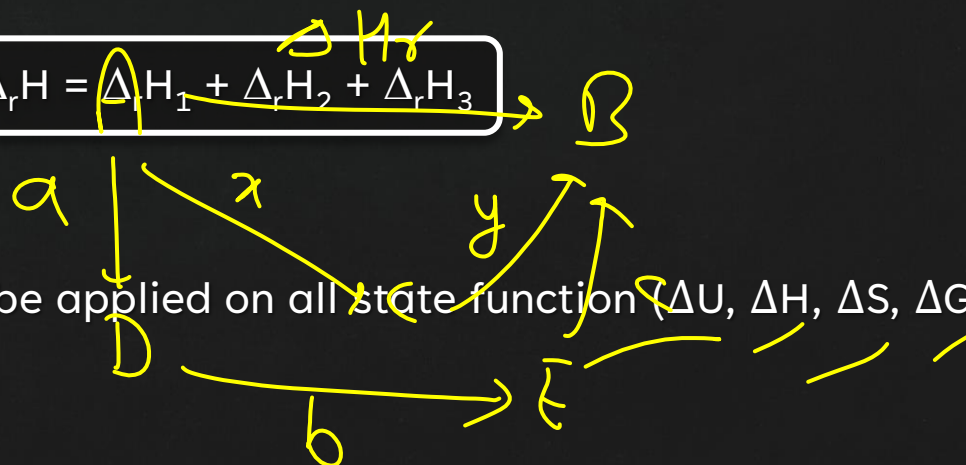
$$\Delta H_r = x + y = a + b + c$$



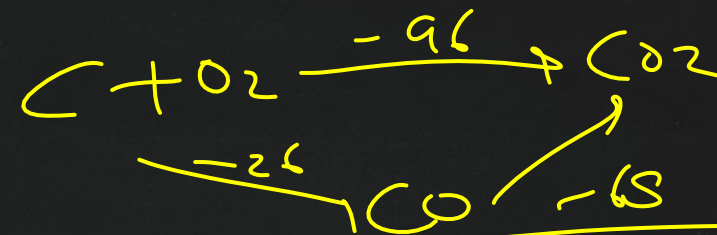
$$\Delta_r H = \Delta_r H_1 + \Delta_r H_2 + \Delta_r H_3$$



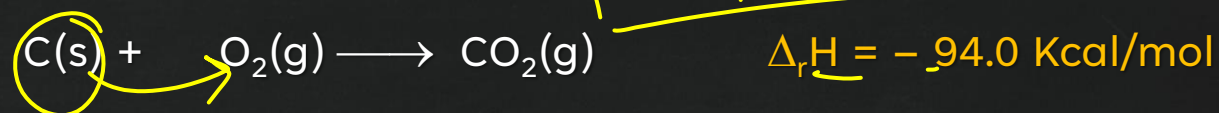
Hess law can be applied on all state function (ΔU , ΔH , ΔS , ΔG)



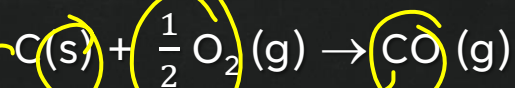
Example



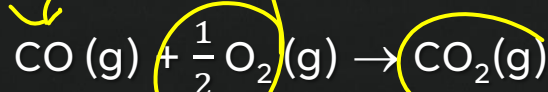
$$\boxed{-96 = -26 - 68 = -94}$$



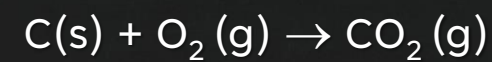
Mechanism



$$\Delta H_1 = -26.0 \text{ kCal/mol}$$



$$\Delta H_2 = -68.0 \text{ kCal/mol}$$



$$\Delta_r H = -26 - 68 = -94.0 \text{ kCal/mol}$$

$$\Delta_r H = \Delta H_1 + \Delta H_2$$

TYPES OF ENTHALPIES OF REACTION



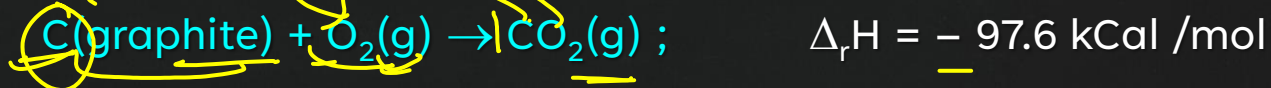
ENTHALPY OF FORMATION ($\Delta_f H$)



Enthalpy change when 1 mole of substance is formed by its constituent elements is called enthalpy of formation.



Species which is being formed is to be taken 1 mole. The entire equation is balanced accordingly.



Here

$$\Delta_r H = (\Delta_f H)_{\text{CO}_2} = -97.6 \text{ Kcal/mol.}$$



Standard enthalpy change for the formation of 1 mole of product from its constituent elements in their most stable states is called standard molar enthalpy of formation ($\Delta_f H^\circ$).

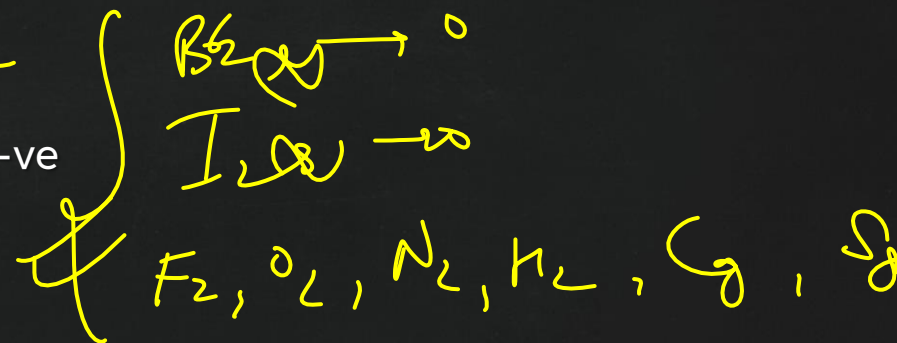


Standard enthalpy of formation of every element in its most stable state or standard state is considered as zero.

$$(\Delta_f H^\circ = 0)$$



$\Delta_f H$ may be +ve or -ve



Application of $\Delta_f H$:

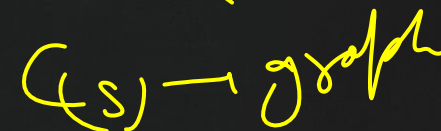
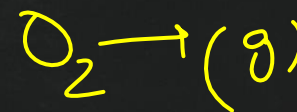
Calculation of $\Delta_r H$ using $\Delta_f H$

! Example

(1) → 1 mole
(2) → constituent, std. state

Which of the following chemical reaction shows the standard enthalpy of formation ?

- (A) $\text{C}(\text{graphite}) + \frac{1}{2} \text{O}_2(\text{g}) \longrightarrow \text{CO}(\text{g})$
- (B) $\text{C}(\text{diamond}) + \text{O}_2(\text{g}) \longrightarrow \text{CO}_2(\text{g})$
- (C) $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \longrightarrow 2\text{HI}(\text{g})$
- (D) $\text{H}_2(\text{g}) + \text{Br}_2(\text{g}) \longrightarrow 2\text{HBr}(\text{g})$

**Solution****Ans : (A)**



ENTHALPY OF COMBUSTION ($\Delta_c H$)

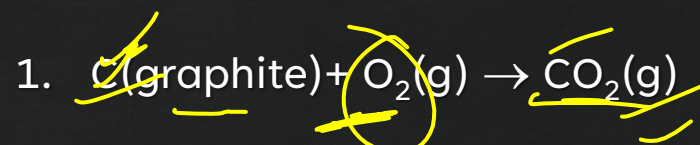


Enthalpy change of reaction, when 1 mole of substance burns in presence of excess oxygen to give products in their most stable state

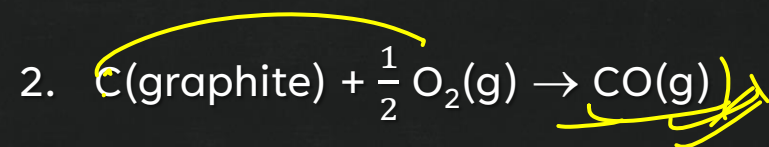


It is denoted by $\Delta_c H$ & is always exothermic in nature.

Example



$$\Delta H = (\Delta_c H)_{\text{carbon}} = (\Delta_f H)_{\text{CO}_2}$$



$$\Delta H \neq (\Delta_c H)_{\text{carbon}}$$

$$\Delta H_r = \Delta H_f(\text{CO}) \neq \Delta H_c(\text{graphite})$$

 NOTE



If organic compounds are given, then compound which contains maximum No. of C—H Bonds will have maximum enthalpy of combustion.

$$\text{Combustion Energy} \propto \text{No. of C - H Bond}$$



Calorific value of Fuel/food : -

The amount of heat produced in calorie or Joule when one gram of a substance (Food & fuel) is completely burnt or oxidised.



1 mole (16gm)

$$(\Delta_c H)_{\text{CH}_4} = -890 \text{ KJ/mole}$$

$$\text{Calorific value} = \frac{\Delta H}{M_{\text{wt}}} = \frac{-890}{16} = -55.6 \text{ kJ/gm}$$

More the C.V., better the fuel.

! Example

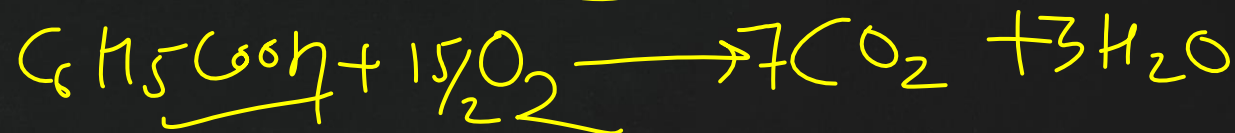
$$\Delta_f H^\circ \text{C}_6\text{H}_5\text{COOH(s)} = -408 \text{ kJ/mol}$$

$$\Delta_f H^\circ \text{H}_2\text{O(l)} = -286 \text{ kJ/mol}$$

$$\Delta_f H^\circ \text{CO}_2(\text{g}) = -393 \text{ kJ/mol}$$

Determine the enthalpy of combustion of $\text{C}_6\text{H}_5\text{COOH(s)}$ at constant volume and at 298 K.

Solution



$$\begin{aligned} \Delta H_r^\circ &= \Delta H_c^\circ(\text{C}_6\text{H}_5\text{COOH}) = 7 \times \Delta H_f^\circ(\text{CO}_2) + 3 \times \Delta H_f^\circ(\text{H}_2\text{O}) \\ &\quad - (\Delta H_f^\circ(\text{C}_6\text{H}_5\text{COOH}) + 15/2 \Delta H_f^\circ(\text{O}_2)) \\ &= -7 \times 393 - 3 \times 286 + 408 \\ &= -2751 - 858 + 408 \end{aligned}$$

Ans: $(\Delta_c H) = -3201 \text{ kJ/mol}$

$$\Delta H = \Delta U + \Delta n_g RT$$



MEASUREMENT OF ΔU & ΔH : CALORIMETRY

BOMB CALORIMETER



Bomb calorimeter is used to find enthalpy of combustion of a fuel/substance.



Bomb calorimeter works on constant volume.



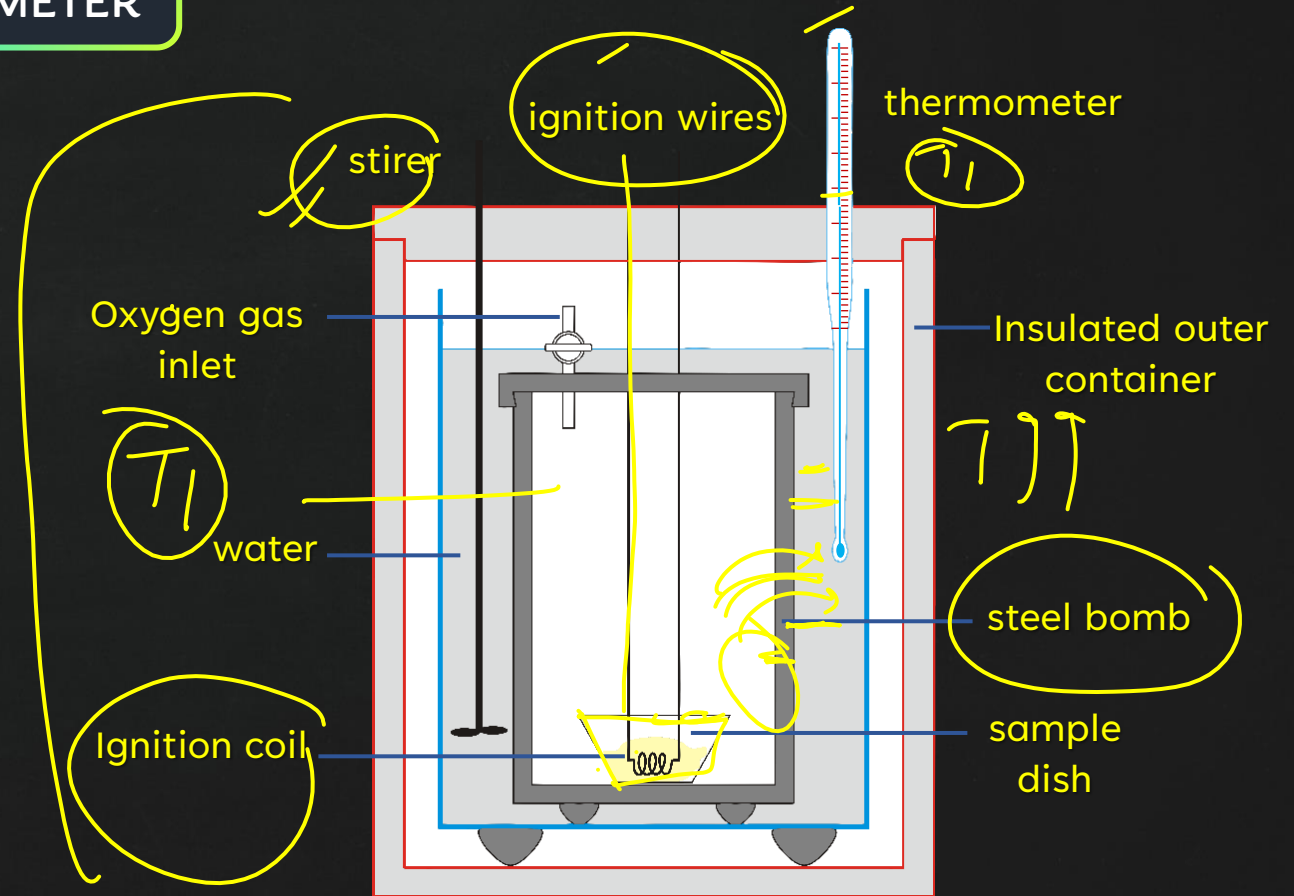
So the heat released is called internal energy.



$$(q_{\text{bomb}})_v = \Delta U$$

total heat = heat by bomb + heat abs by water

BOMB CALORIMETER



T_1 – Temperature before combustion

T_2 – Temperature after combustion

$$\Delta T = T_2 - T_1$$

$$q_b = h C_m \Delta T$$

Heat released q is defined as is following

$$q = C \cdot \Delta T$$

C – total Heat capacity

$$q = n_{\text{Bomb}} C_{m(\text{Bomb})} \Delta T + n_{\text{H}_2\text{O}} C_{m(\text{H}_2\text{O})} \Delta T$$

$$q = m_{\text{Bomb}} C_{S(\text{Bomb})} \Delta T + m_{\text{water}} C_{S(\text{H}_2\text{O})} \Delta T$$

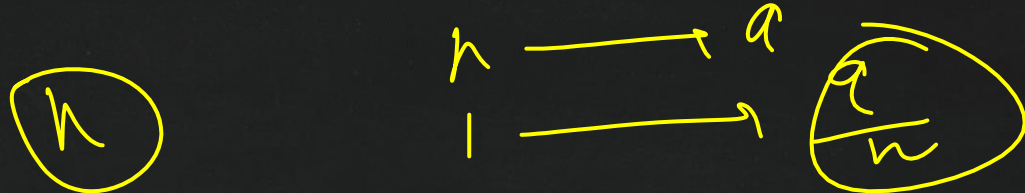
(20)

$\frac{x}{M}$



Let's assume x gm of fuel is taken as sample and molecular mass of fuel is M

$$\text{Then, } n_{\text{fuel}} = \frac{x}{M}$$



 So, n mole of fuel released q amount of heat
 then 1 mole of fuel will release heat = $\frac{q}{n}$

So, $(q_{\text{bomb}})_V = \Delta U = \frac{q}{n}$

Now enthalpy

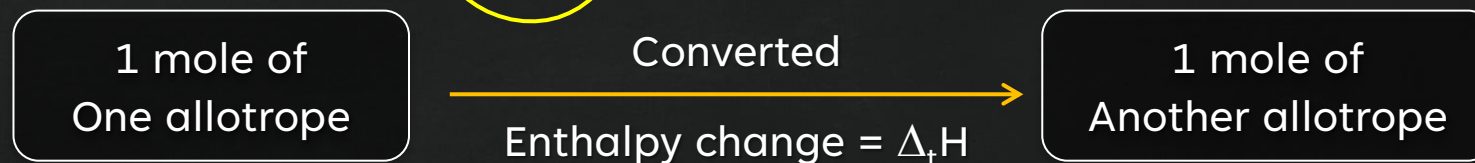
$$\underline{\Delta H} = \Delta U + \Delta n_g RT$$



ENTHALPY OF TRANSITION

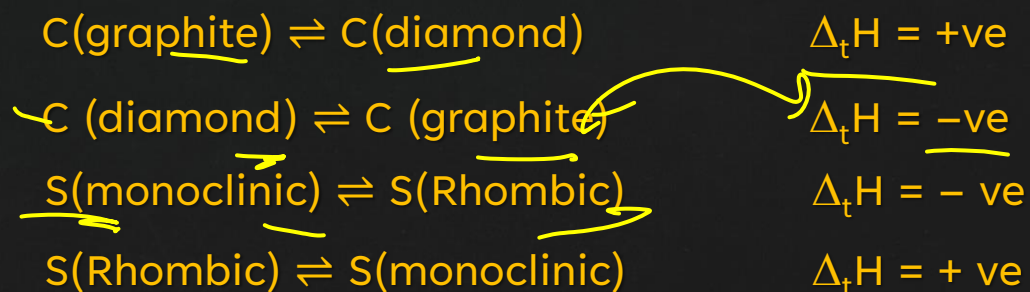


The species undergoing physical state transition is taken 1 mole.



$\Delta_t H$ may be +ve or -ve

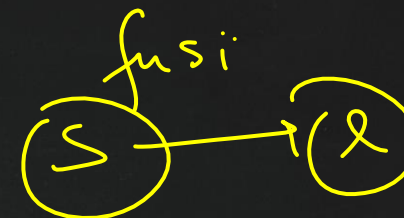
Example



NOTE:- If during any transition, phase of substance is changed then it is also known as **enthalpy of phase transition**.



ENTHALPY OF PHASE TRANSITION



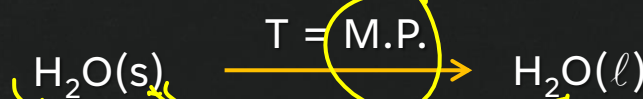
(i) Enthalpy of fusion ($\Delta_{\text{fus}}H$)

1 mole of
Solid substance



1 mole of
Liquid substance

$T, P \rightarrow \text{Constant}$



$\Delta_{\text{fus}}H = +ve$



Latent heat of fusion (L_f)

Amount of heat required when 1 gm of solid substance is converted into 1 gm of liquid at melting point.

$$L_f = \frac{\Delta_{\text{fus}}H}{M_{\text{wt}}}$$

(Q) → (g)

(ii) Enthalpy of vaporization ($\Delta_{\text{vap}}H$)

1 mole of
Liquid

$T \rightarrow \text{Constant}$
 $\Delta_r H = \Delta_{\text{vap}}H$

1 mole of
vapour

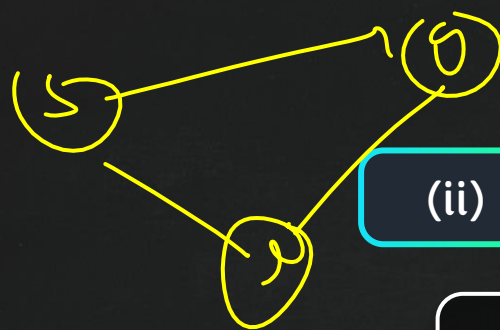
$$\Delta_{\text{vap}}H = +ve$$

Latent heat of vaporization (L_v)



Amount of heat required when 1 gm of liquid substance is converted into 1 gm of vapor at boiling point.

$$L_v = \frac{\Delta_{\text{vap}}H}{M_{\text{wt}}}$$



(ii) Enthalpy of sublimation ($\Delta_{\text{sub}}H$)

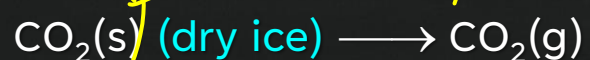
1 mole of
Solid substance

$$\Delta_r H = \Delta_{\text{sub}} H$$

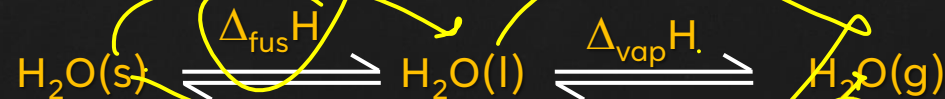
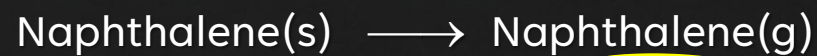
1 mole of
vapour

$$\Delta_{\text{sub}} H = +ve$$

Example



$$\Delta_{\text{sub}} H = 25 \text{ kJ/mol}$$



$$\Delta_{\text{sub}} H = \Delta_{\text{fus}} H + \Delta_{\text{vap}} H$$



BOND ENTHALPY OR BOND ENERGY



Bond enthalpy is the amount of average enthalpy required to break one mole of bonds present in different gaseous covalent compound into free atoms or radicals in the gaseous state.

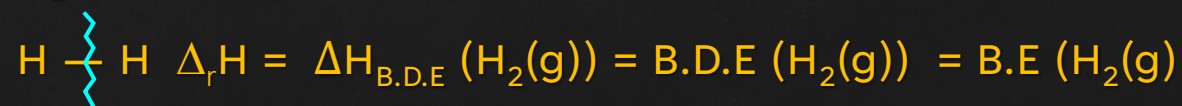


Bond dissociation enthalpy (BDE or $\Delta_{\text{diss}}H$) : -

BDE is the change in enthalpy when 1 mole of covalent bonds of a gaseous diatomic molecule is broken into gaseous atoms or radicals.

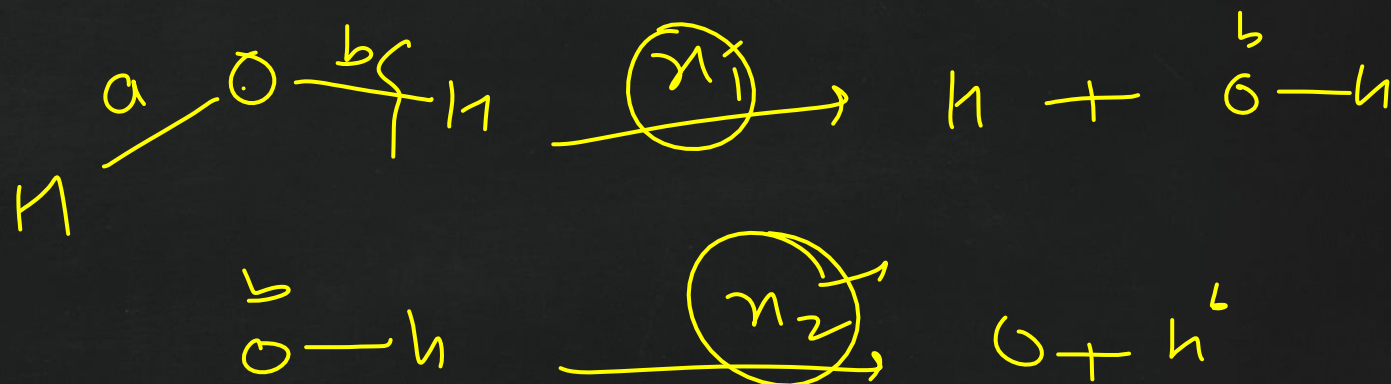


For diatomic molecules B.D.E and Bond enthalpy will be same.

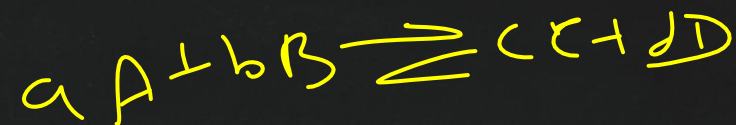




BOND ENTHALPY OR BOND ENERGY



$$\text{B.E.} = \frac{\text{r}_1 + \text{r}_2}{2}$$



APPLICATION OF BOND ENTHALPY ($\Delta_{\text{bond}} H$)

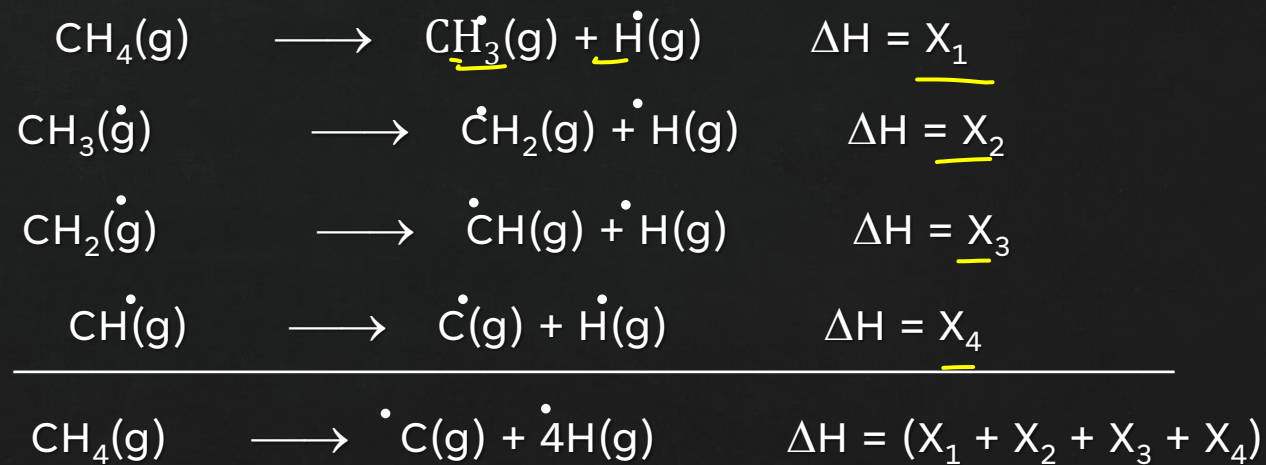
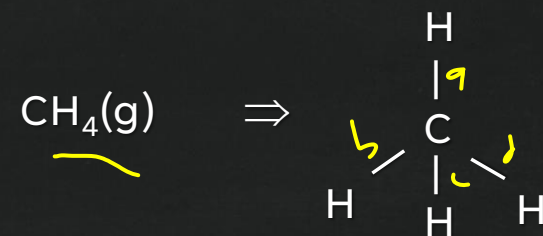
1. Finding enthalpy of Reaction ($\Delta_r H$)

$$\Delta_r H = \sum \Delta_{\text{bond}} H(\text{reactant}) - \sum \Delta_{\text{bond}} H(\text{product})$$

2. If all reactants and products are in standard state, then

$$\Delta_r H^\circ = \sum \text{B. E. (reactant)} - \sum \text{B. E. (Product)}$$

! Example



$$\text{B.E. (C-H)} = \frac{(X_1 + X_2 + X_3 + X_4)}{4}$$



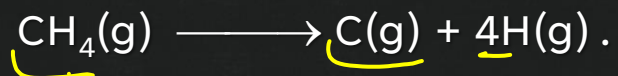
ENTHALPY OF ATOMIZATION ($\Delta_a H$)



Enthalpy change when 1 mole of substance is completely converted into its isolated gaseous atoms by breaking all bonds.



$\Delta_a H$ is always +ve.



$$\underline{\Delta_a H} = \underline{4} \times \underline{(\text{B.E})_{\text{C-H}}}$$



$$\Delta_a H = \frac{1}{2} (\text{B.E})_{\text{H-H}}$$



HEAT OF HYDROGENATION ($\Delta_{\text{HYDROGENATION}} H$)



It is the amount of enthalpy change when 1 mole of unsaturated compound is hydrogenated into saturated compound at a given temperature & pressure.



$$\Delta_r H = \Delta_{\text{Hydrogenation}} H = -ve$$



RESONANCE ENERGY



It is energy difference between resonance hybrid and most stable resonating structure.



Resonance energy is always expressed by negative sign

$$R.E. = [\Delta_{\text{hydrogenation}} H]_{\text{exp}} - [\Delta_{\text{hydrogenation}} H]_{\text{cal.}}$$



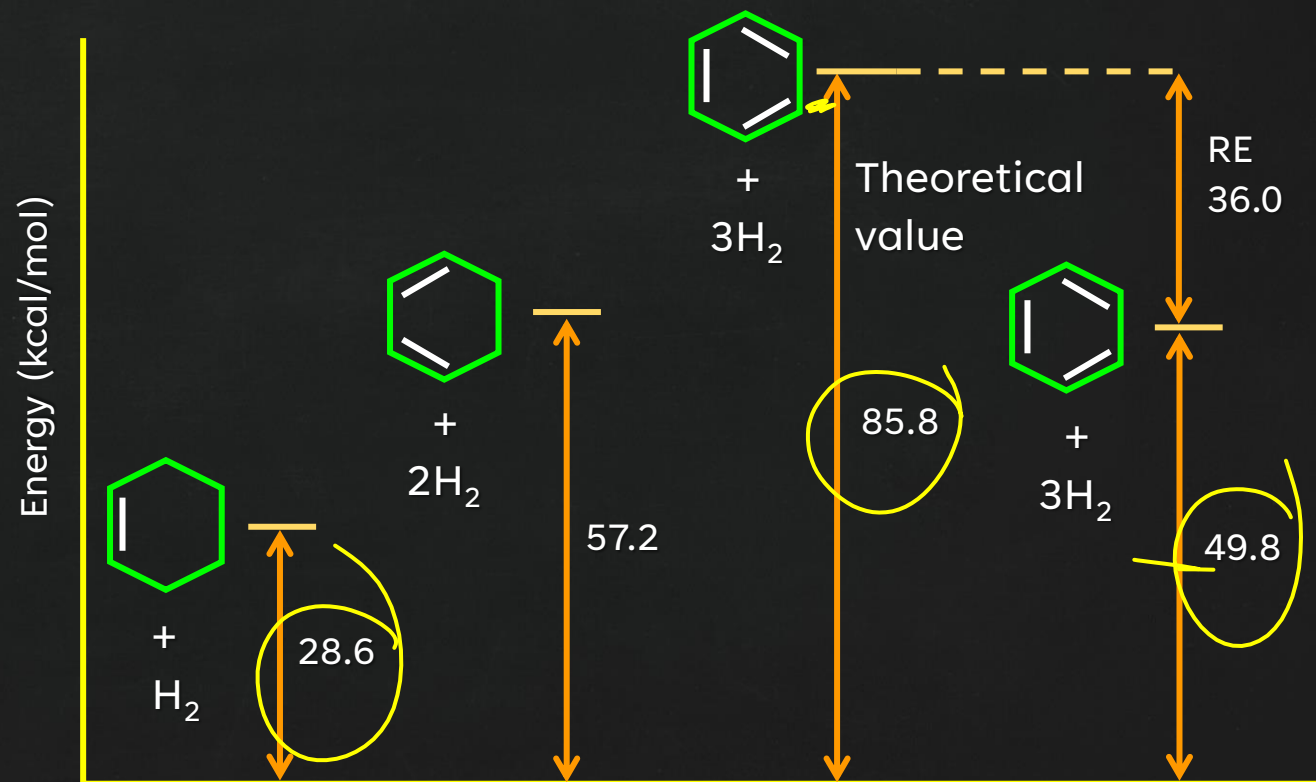
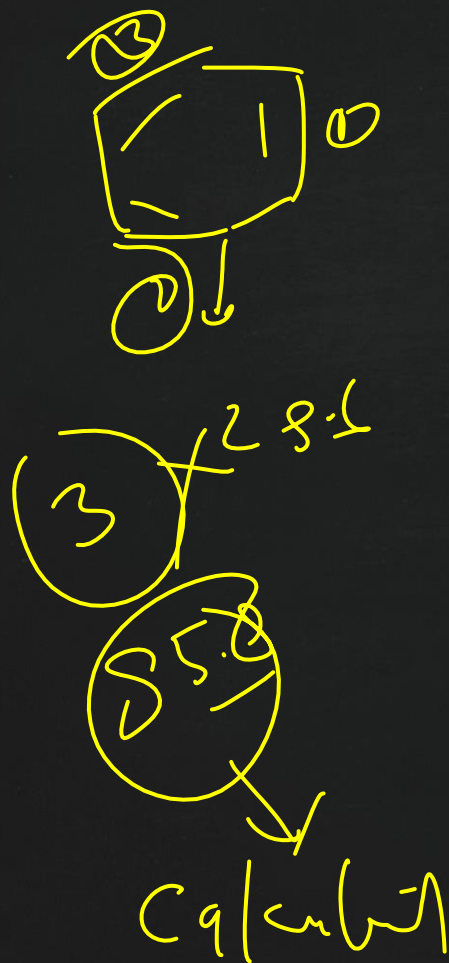
Resonance energy also expressed as

$$R.E. = (\text{Energy})_{\text{reso hybrid}} - (\text{Energy})_{\text{most stable reso structure}}$$

$$R.E. = (\Delta_f H^\circ)_{\text{exp}} - (\Delta_f H^\circ)_{\text{cal.}}$$

$(\Delta_f H^\circ)_{\text{calculated}}$ – it is calculated by bond energy data

$$\Delta_r H^\circ(\text{actual}) - \Delta_r H^\circ(\text{theoretical}) = \sum R.E. (\text{Product}) - \sum R.E. (\text{reactant})$$



$$\text{R.E.} = 49.8 - 85.8 = -36.0 \text{ kcal.}$$



ENTHALPY OF ISOMERISATION

1 mole of
compound

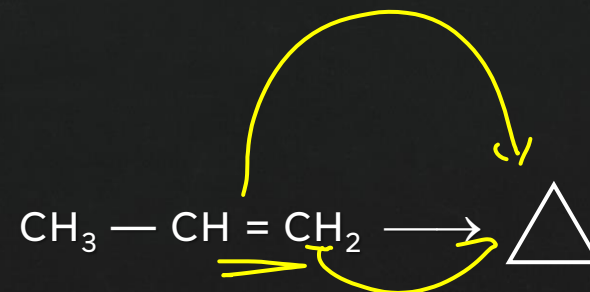
Convert
Enthalpy change = $\Delta_{\text{iso}}H$

1 mole of
another isomer



It may be +ve or -ve

Example



$$\Delta_r H = \Delta_{\text{iso}} H$$



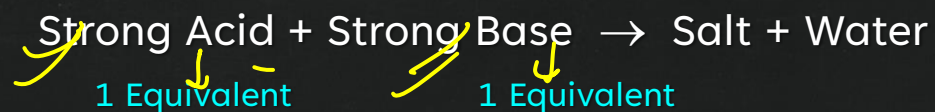
ENTHALPY OF NEUTRALIZATION



Enthalpy change of reaction when 1 gm-equivalent of acid is completely neutralized with 1 gm-equivalent of base.

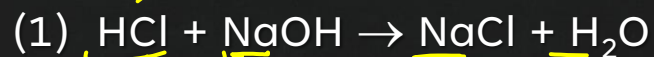


It is always exothermic in nature.

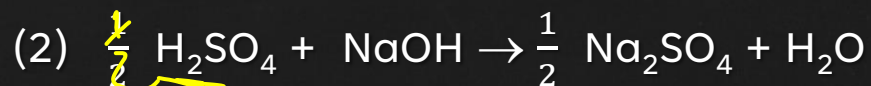


$$\Delta_N H = -13.7 \text{ kCal / Eq.}$$

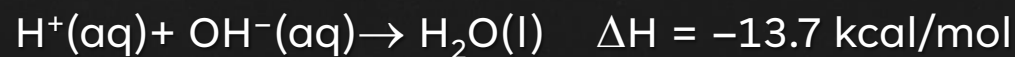
Example



$$\Delta_N H = -13.7 \text{ kcal/mol}$$



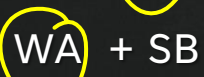
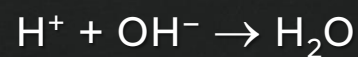
$$\Delta_N H = -13.7 \text{ kcal/eq.}$$



 NOTE



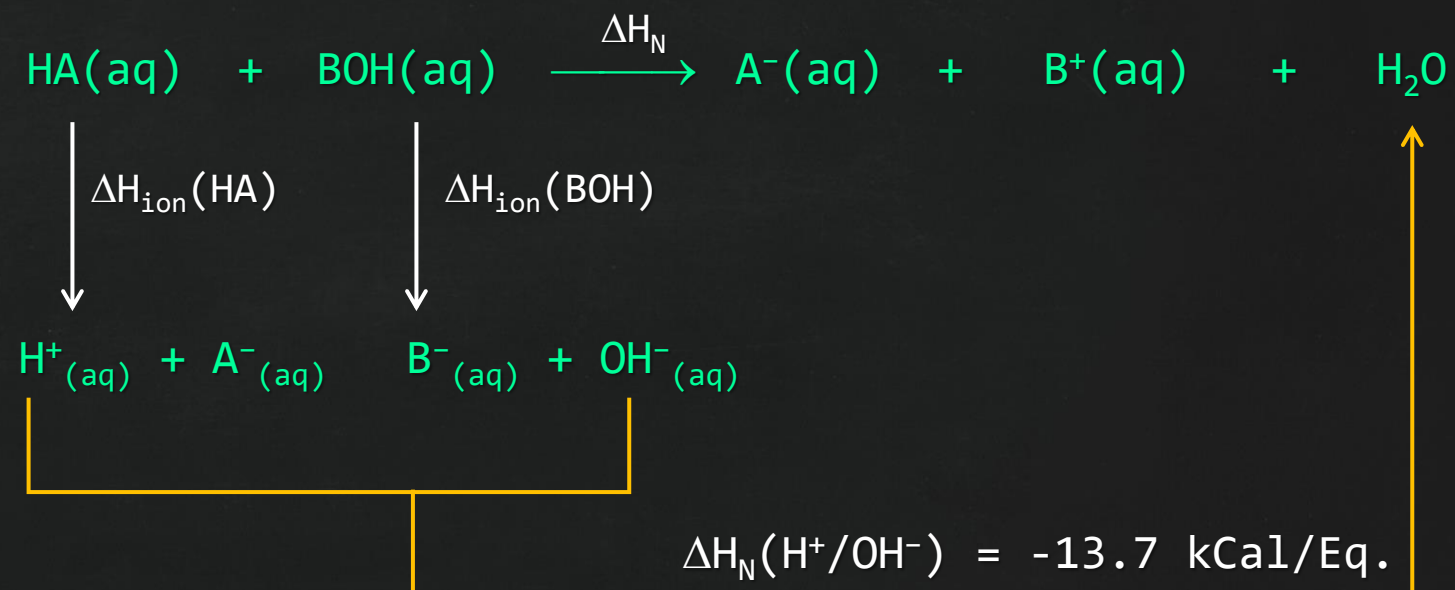
Thus enthalpy of neutralization of SA + SB is equal to enthalpy of reaction :



$\Delta H_N > -13.7 \text{ kCal / Eq.}$

NOTE: Because some part of energy evolved is utilised in the complete ionization of weak acid or weak base or both.

ΔH_N for WA + WB :



So,

$$\Delta H_N(\text{BOH/HA}) = \underbrace{\Delta H_{\text{ion}}(\text{HA}) + \Delta H_{\text{ion}}(\text{BOH})}_{+ve} - 13.7 > -13.7 \text{ kCal/Eq.}$$

Example

1 equivalent of each weak acid HX, HY, HZ is reacted with 1 equivalent of NaOH & enthalpy of neutralization of these acids are -12.0, -8.8 & -10.2 kCal/eq. respectively.

- Find the enthalpy of ionization of all weak acids.
- Arrange all acids in the order of acidic strength.

Solution

$$HX \rightarrow -12.0 - (-13.7) = 1.7$$

$$HY \rightarrow -8.8 + 13.7 = 4.9$$

$$HZ \rightarrow -10.2 + 13.7 = 3.5$$

Ans: 1. $(\Delta_{\text{ion}} H)_{HX} = 1.7 \text{ kCal/eq.}$
 $(\Delta_{\text{ion}} H)_{HY} = 4.9 \text{ kCal/eq.}$
 $(\Delta_{\text{ion}} H)_{HZ} = 3.5 \text{ kCal/eq.}$
 2. $HX > HZ > HY$

$HX > HZ > HY$

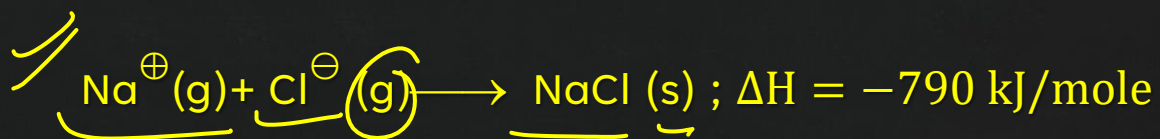


LATTICE ENTHALPY OF CRYSTAL



Lattice enthalpy is defined as amount of heat released when requisite amounts of ions in the gaseous state combine to give 1 mole of crystal lattice.

Example

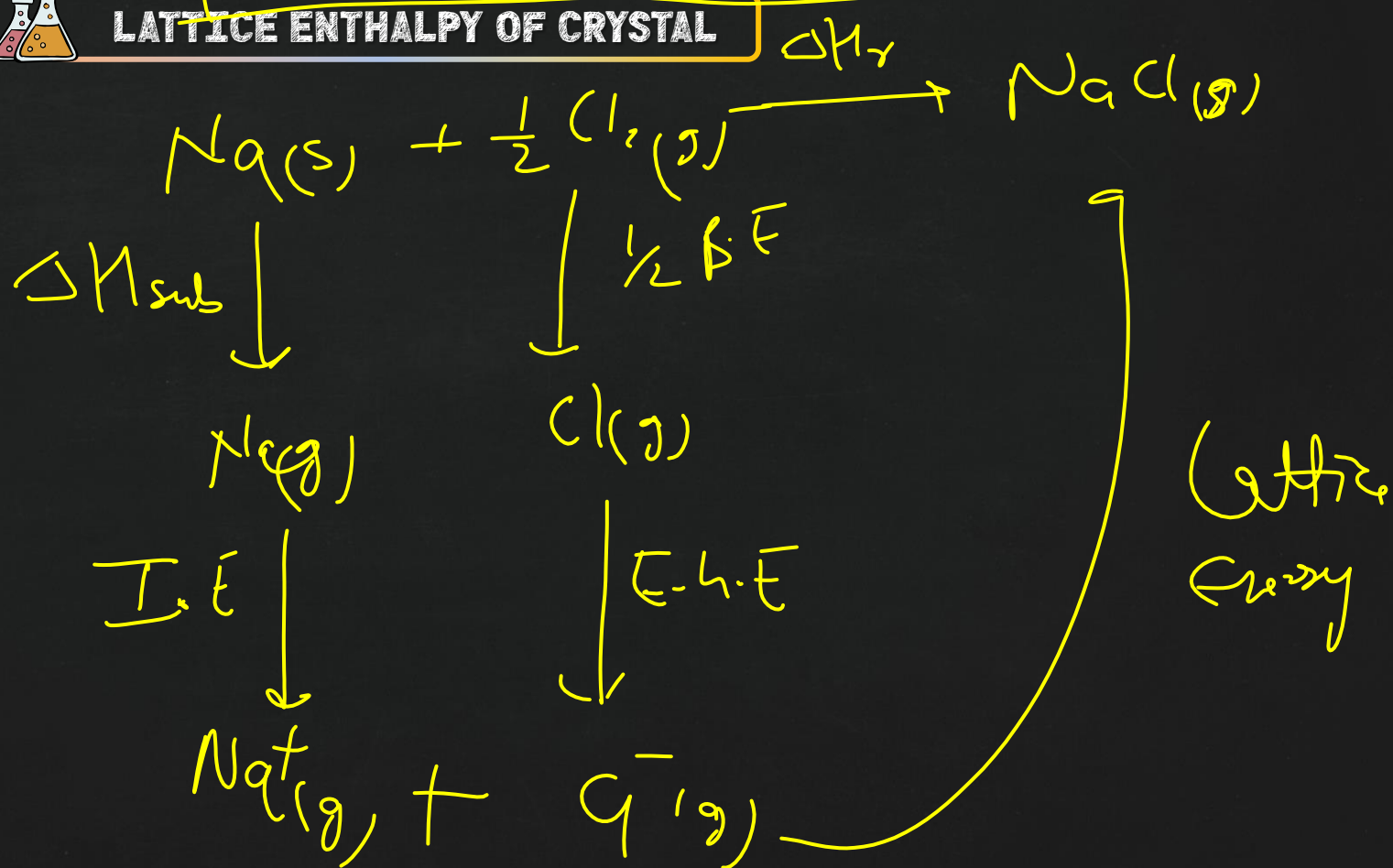


$$\Delta H_f = \Delta H_{sub} + I.E + \frac{1}{2} B.E(u_1) + E.H.E + L.U$$

$$L.U = \Delta H_o - \Delta H_{sub} - I.E - \frac{1}{2} B.E + E.H.E$$



LATTICE ENTHALPY OF CRYSTAL

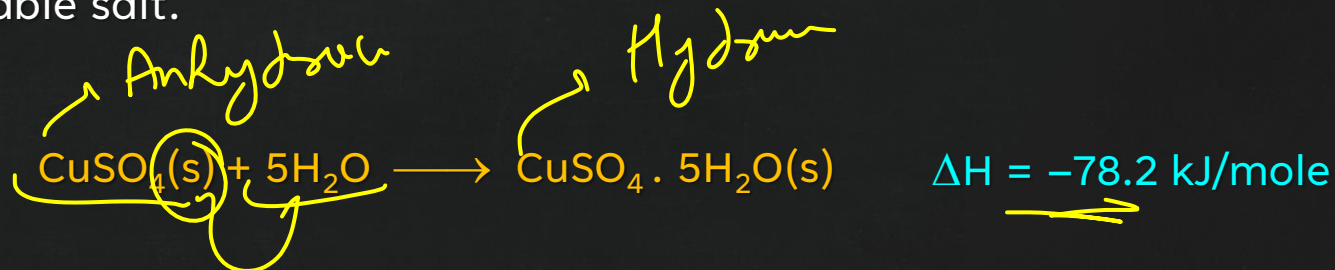




ENTHALPY OF HYDRATION/SOLVATION ($\Delta_{\text{HYD}}H$)



Enthalpy change when 1 mole of anhydrous salt or partially hydrated salt combine with specific number of moles of water to form a new hydrated stable salt.



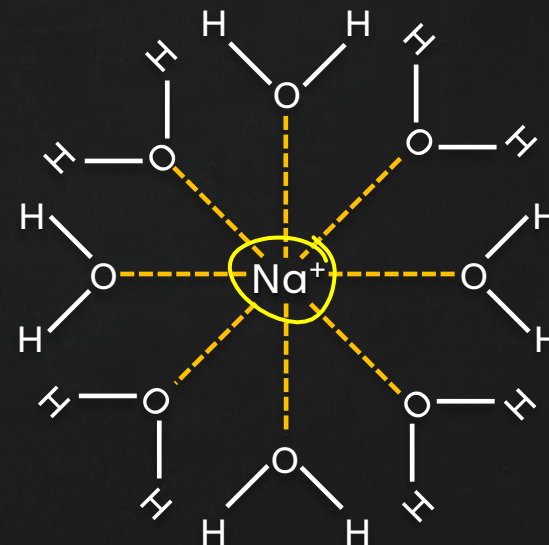
It is exothermic in nature.



ENTHALPY OF HYDRATION/SOLVATION ($\Delta_{\text{HYD}}H$)



If gaseous ion is hydrated, then $\Delta_{\text{Hyd}}H$ is change in enthalpy of 1 mole of gaseous ion surrounded by H_2O .





ENTHALPY OF SOLUTION ($\Delta_{\text{sol}}H$)



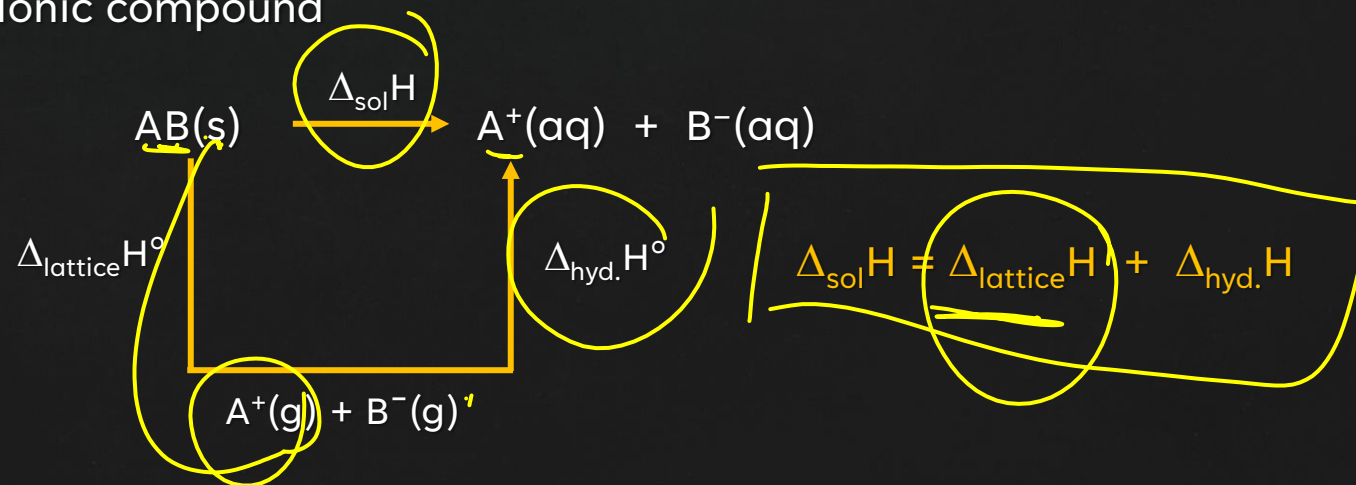
Amount of heat absorbed or evolved when one mole of substance is dissolved in such a large volume of solvent such that further addition of solvent does not produce any more heat change is called as enthalpy of solution or $\Delta_{\text{sol}}H$ at infinite dilution. .

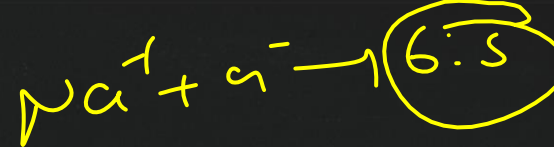
$\Delta_{\text{sol}}H \rightarrow$ Exothermic & Endothermic



NOTE

For Ionic compound





! Example

The lattice energy of solid NaCl is 180 kcal/mole. The dissolution of the solid in water in the form of ions is endothermic to the extent of 1 kcal/mole. If the hydration energies of Na^+ and Cl^- ions are in the ratio 6:5, what is the enthalpy of hydration of sodium ion?

- (A) -85.6 kcal/mole (B) -97.6 kcal/mole
(C) 82.6 kcal/mole (D) none of the above

Solution

$$\Delta H_{sol} = \Delta H_{lattice} + \Delta H_{hyd}$$

$$\therefore \text{Hydration enthalpy} \quad \Delta H_{hyd} = \Delta H_{sol} - \Delta H_{lattice}$$

$$\begin{aligned} \text{of sodium} &= \frac{6}{11} \times -179 = -97.63 \text{ kcal/m} \\ &= 1 \text{ kcal} - 180 = -179 \end{aligned}$$

Ans: (B)

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